



UJI KOMPETENSI KEAHLIAN

Automate web deployment with Ansible on the AWS

Presented by Group 1

Anggota Kelompok



Alifiah Firdiaturrosyidah
(06)



Greenaura Natharina Sunardi
(17)



Muhammad Haidar Abiyyah
Yahya (24)

Deskripsi Proyek dan Tugas

Sebagai seorang spesialis / insinyur TI di sebuah perusahaan besar, Anda bertanggung jawab untuk mengotomatiskan instalasi WordPress dan situs web di server baru.

Perusahaan ingin proses ini diotomatisasi, mulai dari instalasi layanan yang diperlukan hingga konfigurasi WordPress dan situs web, sehingga server siap digunakan dalam waktu sesingkat mungkin.

Kebutuhan Proyek



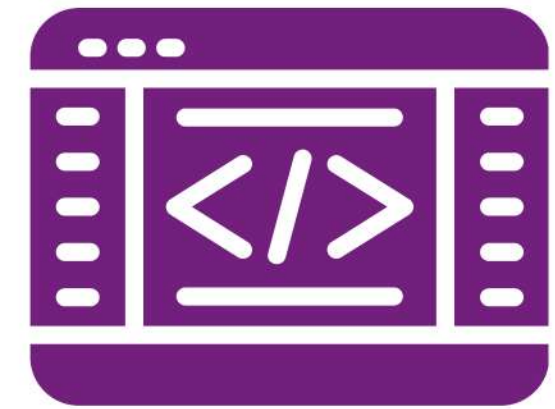
Laptop



Koneksi Internet



Akun AWS Learner
Lab

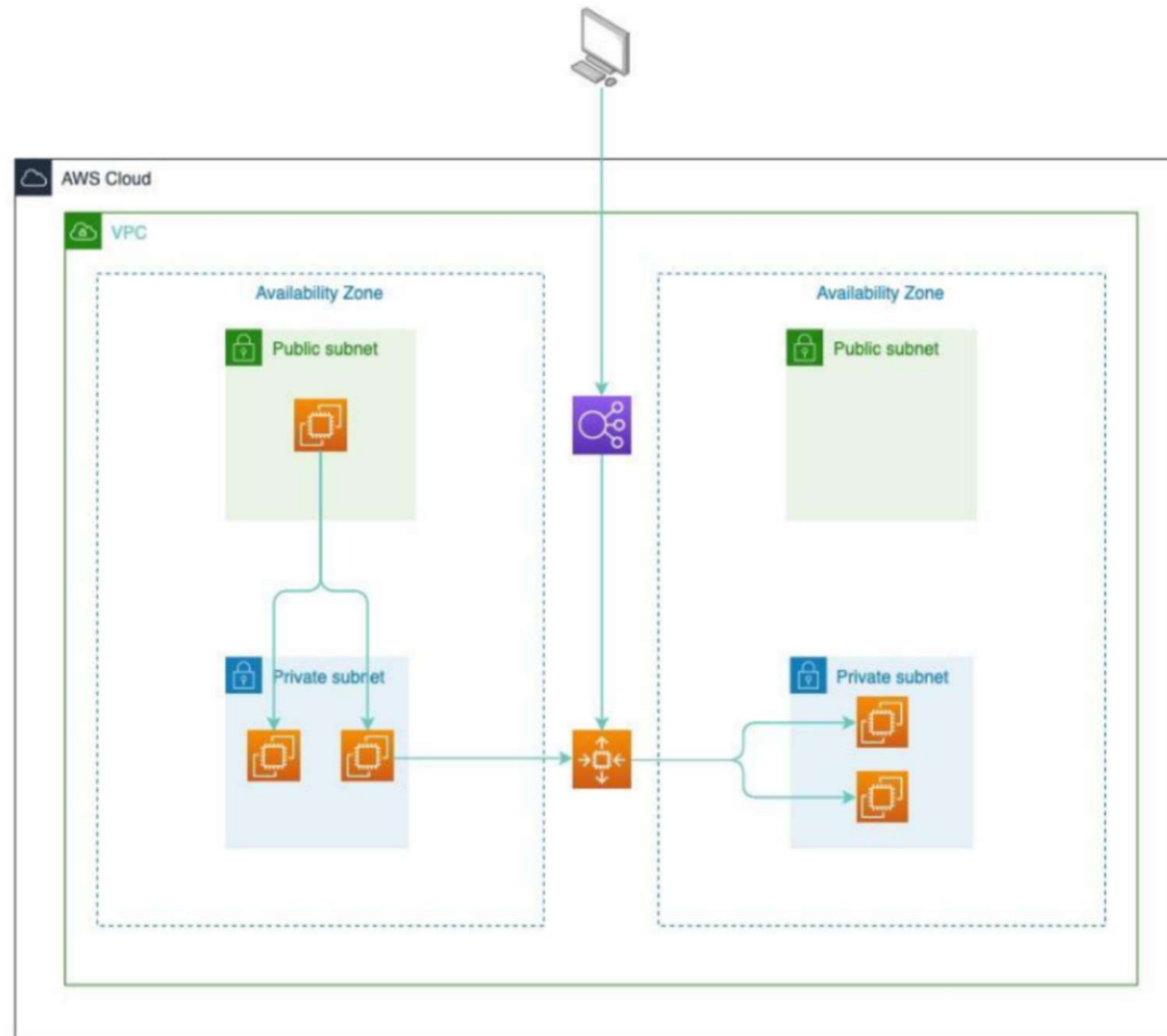


Text Editor

Tugas Proyek

- Buat berkas inventaris untuk server jauh. Pastikan server jauh dapat diakses melalui Node dengan Ansible.
- Buat panduan Ansible untuk mengotomatiskan penginstalan Apache, MySQL, dan PHP, serta mengonfigurasi WordPress dan VulnWeb agar siap digunakan.
- Jalankan panduan dan pastikan WordPress dan VulnWeb telah terinstal dan dapat diakses melalui peramban menggunakan nama DNS Load Balancer.
- Verifikasi bahwa semua konfigurasi telah berjalan dengan benar, dan WordPress serta VulnWeb berfungsi seperti yang diharapkan.
- Untuk VulnWeb, Anda dapat mengunjungi https://github.com/jonisetiyawan48/ukk_tkj_2.git untuk menerapkan aplikasi. Harap baca dengan seksama cara menggunakan aplikasi ini.
- VulnWeb adalah situs web dengan kerentanan keamanan. Tambal kerentanan di situs web.

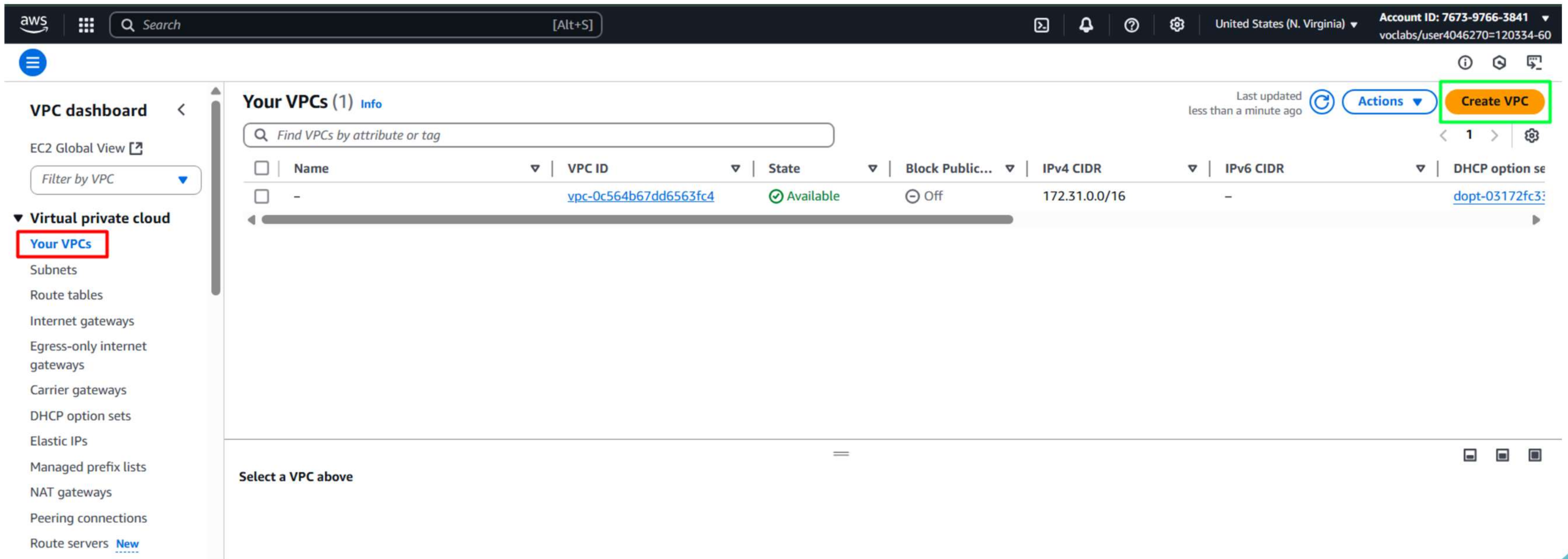
Topologi



Virtual Private Cloud (VPC)

Virtual Private Cloud (VPC)

Membuat jaringan virtual terisolasi di cloud, yang berfungsi memberi kendali penuh atas pengaturan jaringan, keamanan, dan konektivitas antar sumber daya.



The screenshot displays the AWS Management Console interface for the Virtual Private Cloud (VPC) service. The top navigation bar includes the AWS logo, a search bar, and account information for 'United States (N. Virginia)' with Account ID 7673-9766-3841. The left-hand navigation pane is titled 'VPC dashboard' and lists various network resources; 'Your VPCs' is highlighted with a red box. The main content area, titled 'Your VPCs (1)', features a search bar and a table with one VPC entry. The table columns are Name, VPC ID, State, Block Public..., IPv4 CIDR, IPv6 CIDR, and DHCP option set. The single VPC listed has the ID 'vpc-0c564b67dd6563fc4' and is in an 'Available' state. An orange 'Create VPC' button is visible in the top right corner of the main area, highlighted with a green box. Below the table, a message prompts the user to 'Select a VPC above'.

Name	VPC ID	State	Block Public...	IPv4 CIDR	IPv6 CIDR	DHCP option set
-	vpc-0c564b67dd6563fc4	Available	Off	172.31.0.0/16	-	dopt-03172fc3

Setup Konfigurasi VPC

VPC settings

Resources to create [Info](#)

Create only the VPC resource or the VPC and other networking resources.

☐ VPC only

☒ VPC and more

Name tag auto-generation [Info](#)

Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.

☒ Auto-generate

VPC-Group1

IPv4 CIDR block [Info](#)

Determine the starting IP and the size of your VPC using CIDR notation.

192.168.1.0/24

256 IPs

CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)

☒ No IPv6 CIDR block

☐ Amazon-provided IPv6 CIDR block

Tenancy [Info](#)

Default

Number of Availability Zones (AZs) [Info](#)

Choose the number of AZs in which to provision subnets. We recommend at least two AZs for high availability.

1 2 3

► Customize AZs

Number of public subnets [Info](#)

The number of public subnets to add to your VPC. Use public subnets for web applications that need to be publicly accessible over the internet.

0 2

Number of private subnets [Info](#)

The number of private subnets to add to your VPC. Use private subnets to secure backend resources that don't need public access.

0 2 4

▼ Customize subnets CIDR blocks

Public subnet CIDR block in us-east-1a

192.168.1.0/28

16 IPs

Public subnet CIDR block in us-east-1b

192.168.1.16/28

16 IPs

Private subnet CIDR block in us-east-1a

192.168.1.128/28

16 IPs

Private subnet CIDR block in us-east-1b

192.168.1.144/28

16 IPs

NAT gateways (\$) [Info](#)

Choose the number of Availability Zones (AZs) in which to create NAT gateways. Note that there is a charge for each NAT gateway

None

In 1 AZ

1 per AZ

VPC endpoints [Info](#)

Endpoints can help reduce NAT gateway charges and improve security by accessing S3 directly from the VPC. By default, full access policy is used. You can customize this policy at any time.

None

S3 Gateway

DNS options [Info](#)

☒ Enable DNS hostnames

☒ Enable DNS resolution

► Additional tags

Setup Konfigurasi VPC

NAT di VPC memungkinkan subnet privat mengakses internet tanpa membuka akses masuk dari luar, sehingga tetap aman namun tetap terhubung ke luar.

The screenshot displays the AWS Management Console interface for NAT gateways. On the left, the 'VPC dashboard' sidebar is visible, with 'NAT gateways' highlighted under the 'Virtual private cloud' section. The main content area shows the 'NAT gateways' page with a search bar and a table of existing gateways (currently empty). A 'Create NAT gateway' button is highlighted in the top right. A modal window titled 'NAT gateway settings' is open, showing the configuration steps:

- Name - optional:** A text input field containing 'NATGateway-VPC-Group1'.
- Subnet:** A dropdown menu showing 'subnet-0f39347344ba40a87 (VPC-Group1-subnet-public1-us-east-1a)'.
- Connectivity type:** Radio buttons for 'Public' (selected) and 'Private'.
- Elastic IP allocation ID:** A text input field containing 'eipalloc-029d3bea374996c4c'.
- Buttons:** An 'Allocate Elastic IP' button is highlighted in the bottom right of the modal.

Setup Konfigurasi VPC

Route table NAT mengarahkan lalu lintas dari subnet privat ke NAT gateway agar bisa mengakses internet tanpa membuka akses langsung dari luar.

The screenshot displays the AWS Management Console interface for the 'Route tables' section. The left-hand navigation pane shows the 'Route tables' link highlighted with a red rectangular box. The main content area shows a list of route tables with one entry: 'rtb-05745a06c0fd749cd'. The 'Create route table' button in the top right corner is highlighted with a green rectangular box. A modal window titled 'Route table settings' is open, showing the 'Name' field set to 'Route-NATGateway' and the 'VPC' dropdown set to 'vpc-012dfddf7fb5909e3 (VPC-Group1-vpc)'.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

Route-NATGateway

VPC
The VPC to use for this route table.

vpc-012dfddf7fb5909e3 (VPC-Group1-vpc)

Setup Konfigurasi VPC

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (1)

Both

Edit routes

< 1 >

Filter routes

Destination	Target	Status	Propagated
192.168.1.0/24	local	Active	No

Edit routes

Destination

192.168.1.0/24

Target

local

local

NAT Gateway

nat-

Use: "nat-"

nat-0a5247ebf267e884c (NATGateway-VPC-Group1)

Status

Active

-

Propagated

No

No

Remove

Add route

Cancel

Preview

Save changes

Setup Konfigurasi VPC

Routes

Subnet associations

Edge associations

Route propagation

Tags

Explicit subnet associations (0)

Find subnet association

Edit subnet associations

< 1 >

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
No subnet associations You do not have any subnet associations.			

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/4)

Filter subnet associations

< 1 >

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	VPC-Group1-subnet-private1-us-east-1a	subnet-0fc7f8b6696225c45	192.168.1.128/28	–	rtb-08c6dbcdcc4e0d42 / VPC-Group1...
☒	VPC-Group1-subnet-private2-us-east-1b	subnet-03e8d32a359f36cad	192.168.1.144/28	–	rtb-04e98598daf8ef23d / VPC-Group1...
☐	VPC-Group1-subnet-public1-us-east-1a	subnet-0f39347344ba40a87	192.168.1.0/28	–	rtb-02cd6196bb64a1351 / VPC-Group...
☐	VPC-Group1-subnet-public2-us-east-1b	subnet-0c95478eb60b22b5f	192.168.1.16/28	–	rtb-02cd6196bb64a1351 / VPC-Group...

Selected subnets

subnet-0fc7f8b6696225c45 / VPC-Group1-subnet-private1-us-east-1a

subnet-03e8d32a359f36cad / VPC-Group1-subnet-private2-us-east-1b

Cancel

Save associations

Elastic Compute Cloud (EC2)

Elastic Compute Cloud (EC2)

Menyediakan server virtual untuk menjalankan aplikasi, menyimpan data, dan memproses komputasi secara fleksibel di cloud.

The screenshot shows the AWS Management Console interface for the EC2 Instances page. The top navigation bar includes the AWS logo, a search bar, and account information (United States (N. Virginia), Account ID: 7673-9766-3841). The left sidebar shows the EC2 menu with 'Instances' highlighted. The main content area displays the 'Instances' page with a search bar, a filter dropdown set to 'All states', and a table of instances. Since no instances are present, a message states 'No instances. You do not have any instances in this region.' and a 'Launch instances' button is visible. The 'Launch instances' button in the top right corner is highlighted with a green box.

aws | Search [Alt+S] | United States (N. Virginia) | Account ID: 7673-9766-3841 | voclabs/user4046270=120334-60

EC2 > Instances

Instances Info

Last updated less than a minute ago

Connect | Instance state | Actions | **Launch instances**

Find Instance by attribute or tag (case-sensitive) | All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
No instances You do not have any instances in this region								

Launch instances

Konfigurasi EC2

Dalam setup ini, akan dibuat tiga instance EC2 dengan peran berbeda, yaitu:



- **AnsibleNode:** Berfungsi sebagai server kontrol untuk menjalankan dan mengelola otomatisasi menggunakan Ansible.

Konfigurasi EC2

Dalam setup ini, akan dibuat tiga instance EC2 dengan peran berbeda, yaitu:



- **AnsibleNode:** Berfungsi sebagai server kontrol untuk menjalankan dan mengelola otomatisasi menggunakan Ansible.
- **TargetCMS (WordPress):** Instance yang digunakan untuk menghosting aplikasi CMS WordPress.

Konfigurasi EC2

Dalam setup ini, akan dibuat tiga instance EC2 dengan peran berbeda, yaitu:



- **AnsibleNode:** Berfungsi sebagai server kontrol untuk menjalankan dan mengelola otomatisasi menggunakan Ansible.
- **TargetCMS (WordPress):** Instance yang digunakan untuk menghosting aplikasi CMS WordPress.
- **TargetBlog (VulnWeb):** Instance yang digunakan untuk menjalankan aplikasi web rentan sebagai media uji coba atau simulasi keamanan.

Konfigurasi EC2 (AnsibleNode)

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Quick Start



[Browse more AMIs](#)
Including AMIs from
AWS, Marketplace and
the Community

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type
ami-084568db4383264d4 (64-bit (x86)) / ami-0c4e709339fa8521a (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

Konfigurasi EC2 (AnsibleNode)

Description

Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 24.04, amd64 noble image

Architecture

64-bit (x86)

AMI ID

ami-084568db4383264d4

Publish Date

2025-03-05

Username

ubuntu



Verified provider

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t2.micro

Free tier eligible

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Windows base pricing: 0.0162 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour

On-Demand SUSE base pricing: 0.0116 USD per Hour On-Demand RHEL base pricing: 0.026 USD per Hour

On-Demand Linux base pricing: 0.0116 USD per Hour

☒ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

vockey



[Create new key pair](#)

Konfigurasi EC2 (AnsibleNode)

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-012dfddf7fb5909e3 (VPC-Group1-vpc)
192.168.1.0/24



Subnet [Info](#)

subnet-0f39347344ba40a87
VPC: vpc-012dfddf7fb5909e3 Owner: 767397663841 Availability Zone: us-east-1a
Zone type: Availability Zone IP addresses available: 10 CIDR: 192.168.1.0/28

VPC-Group1-subnet-public1-us-east-1a



[Create new subnet](#)

Auto-assign public IP [Info](#)

Enable

[Additional charges apply](#) when outside of [free tier allowance](#)

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

Security group name - required

SG-AnsibleNode

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and . _ - / () # , @ [] + = & ; { } ! \$ *

Description - required [Info](#)

Allow SSH access only

Konfigurasi EC2 (AnsibleNode)

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0) Remove

Type Info	Protocol Info	Port range Info
ssh ▼	TCP	22
Source type Info	Source Info	Description - optional Info
Anywhere ▼	Add CIDR, prefix list or security group 0.0.0.0/0 ✕	e.g. SSH for admin desktop

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only. ✕

Add security group rule

► Advanced network configuration

▼ Configure storage Info Advanced

1x GiB ▼ Root volume, 3000 IOPS, Not encrypted

Info Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage ✕

Add new volume

Konfigurasi EC2 (TargetCMS)

Instances (1/1) [Info](#)

Last updated 3 minutes ago [Refresh](#) [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

[All states](#)

☒	Name ✎	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public
☒	AnsibleNode	i-00f55a6c0947417a7	Running	t2.micro	Initializing	View alarms +	us-east-1a	44.201

[Connect](#)
[View details](#)
[Manage instance state](#)
[Instance settings](#)
[Networking](#)
[Security](#)
[Image and templates](#)
[Monitor and troubleshoot](#)

[Create image](#)
[Create template from instance](#)
[Launch more like this](#)

Konfigurasi EC2 (TargetCMS)

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below



[AMI from catalog](#)

[Recents](#)

[Quick Start](#)

Name

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type

Verified provider

Free tier eligible

Description

Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 24.04, amd64 noble image

Image ID

ami-084568db4383264d4

Username 



[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

Konfigurasi EC2 (TargetCMS)

Username ⓘ

ubuntu

Published

2025-03-05T09:18:37.000Z

Architecture

x86_64

Virtualization

hvm

Root device type

ebs

ENA Enabled

Yes

Boot mode

uefi-preferred

▼ Instance type ⓘ | Get advice

Instance type

t2.micro

Free tier eligible

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Windows base pricing: 0.0162 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour

On-Demand SUSE base pricing: 0.0116 USD per Hour On-Demand RHEL base pricing: 0.026 USD per Hour

On-Demand Linux base pricing: 0.0116 USD per Hour

☐ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) ⓘ

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

vockey

▼

↻

Create new key pair

Konfigurasi EC2 (TargetCMS)

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-012dfddf7fb5909e3 (VPC-Group1-vpc)
192.168.1.0/24



Subnet [Info](#)

subnet-0fc7f8b6696225c45 **VPC-Group1-subnet-private1-us-east-1a**
VPC: vpc-012dfddf7fb5909e3 Owner: 767397663841 Availability Zone: us-east-1a
Zone type: Availability Zone IP addresses available: 10 CIDR: 192.168.1.128/28



[Create new subnet](#)

Auto-assign public IP [Info](#)

Disable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

Security group name - required

SG-TargetCMS

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and . _ - / () # , @ [] + = & ; { } ! \$ *

Description - required [Info](#)

Allow SSH and HTTP access

Konfigurasi EC2 (TargetCMS)

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0)

Type | Info

ssh ▼

Source type | Info

Anywhere ▼

Protocol | Info

TCP

Port range | Info

22

Source | Info

Q Add CIDR, prefix list or security group

0.0.0.0/0 X

Description - optional | Info

e.g. SSH for admin desktop

Remove

▼ Security group rule 2 (TCP, 80, 0.0.0.0/0)

Type | Info

HTTP ▼

Source type | Info

Anywhere ▼

Protocol | Info

TCP

Port range | Info

80

Source | Info

Q Add CIDR, prefix list or security group

0.0.0.0/0 X

Description - optional | Info

e.g. SSH for admin desktop

Remove

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

X

Add security group rule

► Advanced network configuration

Konfigurasi EC2

Selanjutnya, kita akan masuk atau terhubung ke instance AnsibleNode untuk melakukan konfigurasi dan deployment web pada TargetCMS dan TargetBlog. Namun, sebelum itu pastikan terlebih dahulu bahwa **instance sudah dalam keadaan running** dan **status check menunjukkan 2/2 checks passed** agar siap digunakan.

Instances (1/3) [Info](#)

Last updated 2 minutes ago [Refresh](#) [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

[All states](#)

☐	Name 🔗	Instance ID	Instance state ▼	Instance type ▼	Status check	Alarm status	Availability Zone ▼	Public IPv4 DNS ▼	Public IP ▼
☒	AnsibleNode	i-00f55a6c0947417a7	Running 🔍 🔍	t2.micro	2/2 checks passed	View alarms +	us-east-1a	ec2-44-201-190-154.co...	44.20
☐	TargetBlog	i-0981f2867ef76384c	Running 🔍 🔍	t2.micro	2/2 checks passed	View alarms +	us-east-1a	-	-
☐	TargetCMS	i-04e877d36e99d8e58	Running 🔍 🔍	t2.micro	2/2 checks passed	View alarms +	us-east-1a	-	-

Konfigurasi EC2

Connect to instance [Info](#)

Connect to your instance i-00f55a6c0947417a7 (AnsibleNode) using any of these options

[EC2 Instance Connect](#) | [Session Manager](#) | [SSH client](#) | [EC2 serial console](#)

Instance ID
i-00f55a6c0947417a7 (AnsibleNode)

1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is vockey.pem
3. Run this command, if necessary, to ensure your key is not publicly viewable.
chmod 400 "vockey.pem"
4. Connect to your instance using its Public DNS:
ec2-44-201-190-154.compute-1.amazonaws.com

Example:
ssh -i "vockey.pem" ubuntu@ec2-44-201-190-154.compute-1.amazonaws.com

Note: In most cases, the guessed username is correct. However, read your AMI usage

aws academy

Account

Dashboard

Courses

Calendar

Inbox

History

Help

ALLv2EN-US-LTI1... > Modules > AWS Academy Le... > Launch AWS Academy Learner Lab

AWS

Used \$1 of \$50

02:12

Start Lab

End Lab

AWS Details

Readme

Reset

Close

Cloud Access

AWS CLI: Show

Cloud Labs

Remaining session time: 02:11:55(132 minutes)

Session started at: 2025-05-01T16:08:56-0700

Session to end at: 2025-05-01T20:08:56-0700

Accumulated lab time: 07:20:00 (440 minutes)

No running instance

SSH key Show Download PEM Download PPK

AWS SSO Download URL

AWSAccountId 767397663841

Region us-east-1

Previous

Next

Konfigurasi EC2

Setelah terhubung ke AnsibleNode, langkah pertama yang dilakukan adalah **memperbarui sistem** menggunakan perintah **update** dan **upgrade**, lalu menginstal Ansible agar siap digunakan untuk proses otomatisasi.

```
ubuntu@ip-192-168-1-7: ~  
PS C:\Users\moklet\Downloads> ssh -i "vockey.pem" ubuntu@ec2-44-201-190-154.compute-1.amazonaws.com  
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.8.0-1024-aws x86_64)
```

```
ubuntu@ip-192-168-1-7:~$ sudo su  
root@ip-192-168-1-7:/home/ubuntu#
```

```
root@ip-192-168-1-7:/home/ubuntu# apt update && apt upgrade -y && apt install ansible -y  
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease  
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]  
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]  
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]  
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
```

```
root@ip-192-168-1-7:/home/ubuntu# ansible --version  
ansible [core 2.16.3]  
  config file = None  
  configured module search path = ['/root/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']  
  ansible python module location = /usr/lib/python3/dist-packages/ansible  
  ansible collection location = /root/.ansible/collections:/usr/share/ansible/collections  
  executable location = /usr/bin/ansible
```

Konfigurasi EC2

Sebelum membuat atau memakai inventory.ini, lakukan ***chmod 400 vockey.pem*** agar file key bisa digunakan Ansible tanpa error saat koneksi SSH.

```
root@ip-192-168-1-7: /home/u x Windows PowerShell x + v
PS C:\Users\moklet\Downloads> scp -i "vockey.pem" vockey.pem ubuntu@ec2-44-201-190-154.compute-1.amazonaws.com:/home/ubuntu
vockey.pem 100% 1678 6.4KB/s 00:00
PS C:\Users\moklet\Downloads> |
```

```
root@ip-192-168-1-7:/home/ubuntu# ll
total 36
drwxr-x--- 4 ubuntu ubuntu 4096 May  2 01:19 ./
drwxr-xr-x 3 root    root    4096 May  1 23:48 ../
-rw-r--r-- 1 ubuntu ubuntu  220 Mar 31  2024 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Mar 31  2024 .bashrc
drwx----- 2 ubuntu ubuntu 4096 May  2 01:00 .cache/
-rw-r--r-- 1 ubuntu ubuntu  807 Mar 31  2024 .profile
drwx----- 2 ubuntu ubuntu 4096 May  1 23:48 .ssh/
-rw-r--r-- 1 ubuntu ubuntu    0 May  2 01:01 .sudo_as_admin_successful
-rw-r--r-- 1 root    root    172 May  2 01:12 inventory.ini
-rw-rw-r-- 1 ubuntu ubuntu 1678 May  2 01:19 vockey.pem
```

```
root@ip-192-168-1-7:/home/ubuntu# chmod 400 vockey.pem
root@ip-192-168-1-7:/home/ubuntu# ll
total 36
drwxr-x--- 4 ubuntu ubuntu 4096 May  2 01:19 ./
drwxr-xr-x 3 root    root    4096 May  1 23:48 ../
-rw-r--r-- 1 ubuntu ubuntu  220 Mar 31  2024 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Mar 31  2024 .bashrc
drwx----- 2 ubuntu ubuntu 4096 May  2 01:00 .cache/
-rw-r--r-- 1 ubuntu ubuntu  807 Mar 31  2024 .profile
drwx----- 2 ubuntu ubuntu 4096 May  1 23:48 .ssh/
-rw-r--r-- 1 ubuntu ubuntu    0 May  2 01:01 .sudo_as_admin_successful
-rw-r--r-- 1 root    root    172 May  2 01:12 inventory.ini
-r----- 1 ubuntu ubuntu 1678 May  2 01:19 vockey.pem
```

Konfigurasi EC2

inventory.ini adalah file Ansible yang berisi daftar server target beserta detail koneksi SSH-nya, digunakan agar Ansible dapat mengakses dan mengelola host secara otomatis.

```
root@ip-192-168-1-7:/home/ubuntu# nano inventory.ini
```

i-04e877d36e99d8e58 (TargetCMS)

[Details](#) | [Status and alarms](#) | [Monitoring](#) | [Security](#) | [Networking](#) | [Storage](#) | [Tags](#)

▼ Instance summary [Info](#)

Instance ID
i-04e877d36e99d8e58

IPv6 address
-

Public IPv4 address
-

Instance state
✓ Running

Private IPv4 addresses
192.168.1.140

Public IPv4 DNS
-

IP TargetCMS (WordPress)

i-0981f2867ef76384c (TargetBlog)

[Details](#) | [Status and alarms](#) | [Monitoring](#) | [Security](#) | [Networking](#) | [Storage](#) | [Tags](#)

▼ Instance summary [Info](#)

Instance ID
i-0981f2867ef76384c

IPv6 address
-

Public IPv4 address
-

Instance state
✓ Running

Private IPv4 addresses
192.168.1.142

Public IPv4 DNS
-

IP TargetBlog (VulnWeb)

Konfigurasi EC2

```
GNU nano 7.2 inventory.ini
[wordpress]
192.168.1.140 ansible_user=ubuntu ansible_ssh_private_key_file=./vockey.pem
[blog]
192.168.1.142 ansible_user=ubuntu ansible_ssh_private_key_file=./vockey.pem
```

- **[host_group]:** Menunjukkan nama grup host. Bisa diberi nama sesuai fungsinya, misalnya [wordpress] atau [blog].
- **your_host_ip:** Alamat IP publik dari instance yang akan dikontrol oleh Ansible.
- **ansible_user:** Nama user yang digunakan untuk login ke server via SSH (contohnya: ubuntu untuk Ubuntu AMI).
- **ansible_ssh_private_key_file:** Path ke file private key (.pem) yang digunakan untuk otentikasi SSH.

Konfigurasi EC2

Command ansible *-i <inventory_file> <host_group> -m ping* digunakan untuk mengecek koneksi antara Ansible dan host yang ditentukan dalam file inventory, memastikan bahwa host dapat dijangkau dan siap dikelola oleh Ansible.

```
root@ip-192-168-1-7:/home/ubuntu# ansible -i inventory.ini wordpress -m ping
The authenticity of host '192.168.1.140 (192.168.1.140)' can't be established.
ED25519 key fingerprint is SHA256:9n0tDRUsWQ1Tf3Pui0o3XEPDpBhcGHof8s3QqzSeFPY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
192.168.1.140 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
```

```
root@ip-192-168-1-7:/home/ubuntu# ansible -i inventory.ini blog -m ping
The authenticity of host '192.168.1.142 (192.168.1.142)' can't be established.
ED25519 key fingerprint is SHA256:bRjTEHyKD+N4zzxBGi2dxa5DA5p+1V4p6rknbP4mZy8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
192.168.1.142 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
```

Konfigurasi EC2

"Setelah memastikan koneksi berhasil, langkah selanjutnya adalah membuat file *playbook.yml*, yaitu file berisi sekumpulan task atau instruksi otomatisasi yang akan dijalankan Ansible pada host target sesuai kebutuhan.

```
root@ip-192-168-1-7:/home/ubuntu# nano wordpress.yml
```

```
- hosts: wordpress|
  become: yes
  vars:
    wp_db_name: wordpress
    wp_user: wordpress
    wp_user_pass: wordpress
    wp_root_pass: 123wordpress
    web_root: /var/www/wordpress

  tasks:
    - name: Update package repository
      apt:
        update_cache: yes

    - name: Install Apache, MySQL, and PHP
      apt:
        name:
          - apache2
          - mysql-server
          - php
          - php-mysql
        state: present
```

```
root@ip-192-168-1-7:/home/ubuntu# nano blog.yml
```

```
- hosts: blog
  become: yes
  vars:
    mysql_db_name: cyber23
    mysql_user: cyber
    mysql_user_pass: pass2023
    mysql_root_pass: 19091Cyber
    git_repo: "https://github.com/jonisetiawan48/ukk_tkj_2.git"
    web_root: "/var/www/html"

  tasks:
    - name: Update package repository
      apt:
        update_cache: yes

    - name: Install Apache, MariaDB, and PHP
      apt:
        name:
          - apache2
          - mariadb-server
          - php
          - php-mysqli
          - php-mbstring
        state: present
```

Konfigurasi EC2

Setelah file playbook dibuat, jalankan command *ansible-playbook -i inventory.ini playbook.yml* untuk menjalankan otomatisasi task yang terdapat dalam file playbook ke host target yang didefinisikan di file inventory.

```
root@ip-192-168-1-7:/home/ubuntu# ansible-playbook -i inventory.ini wordpress.yml
```

```
root@ip-192-168-1-7:/home/ubuntu# ansible-playbook -i inventory.ini blog.yml
```

```
TASK [Enable Apache mod_rewrite] *****
changed: [192.168.1.140]

TASK [Restart Apache to apply changes] *****
changed: [192.168.1.140]

RUNNING HANDLER [Restart Apache] *****
changed: [192.168.1.140]

PLAY RECAP *****
192.168.1.140      : ok=15  changed=14  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
```

```
TASK [Import SQL file into database] *****
changed: [192.168.1.142]

TASK [Enable the new site configuration] *****
changed: [192.168.1.142]

TASK [Restart Apache to apply changes] *****
changed: [192.168.1.142]

PLAY RECAP *****
192.168.1.142      : ok=21  changed=13  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
```

Konfigurasi EC2

Setelah file playbook dibuat, jalankan command *ansible-playbook -i inventory.ini playbook.yml* untuk menjalankan otomatisasi task yang terdapat dalam file playbook ke host target yang didefinisikan di file inventory.

```
root@ip-192-168-1-7:/home/ubuntu# ansible-playbook -i inventory.ini wordpress.yml
```

```
root@ip-192-168-1-7:/home/ubuntu# ansible-playbook -i inventory.ini blog.yml
```

```
TASK [Enable Apache mod_rewrite] *****
changed: [192.168.1.140]

TASK [Restart Apache to apply changes] *****
changed: [192.168.1.140]

RUNNING HANDLER [Restart Apache] *****
changed: [192.168.1.140]

PLAY RECAP *****
192.168.1.140      : ok=15  changed=14  unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

```
TASK [Import SQL file into database] *****
changed: [192.168.1.142]

TASK [Enable the new site configuration] *****
changed: [192.168.1.142]

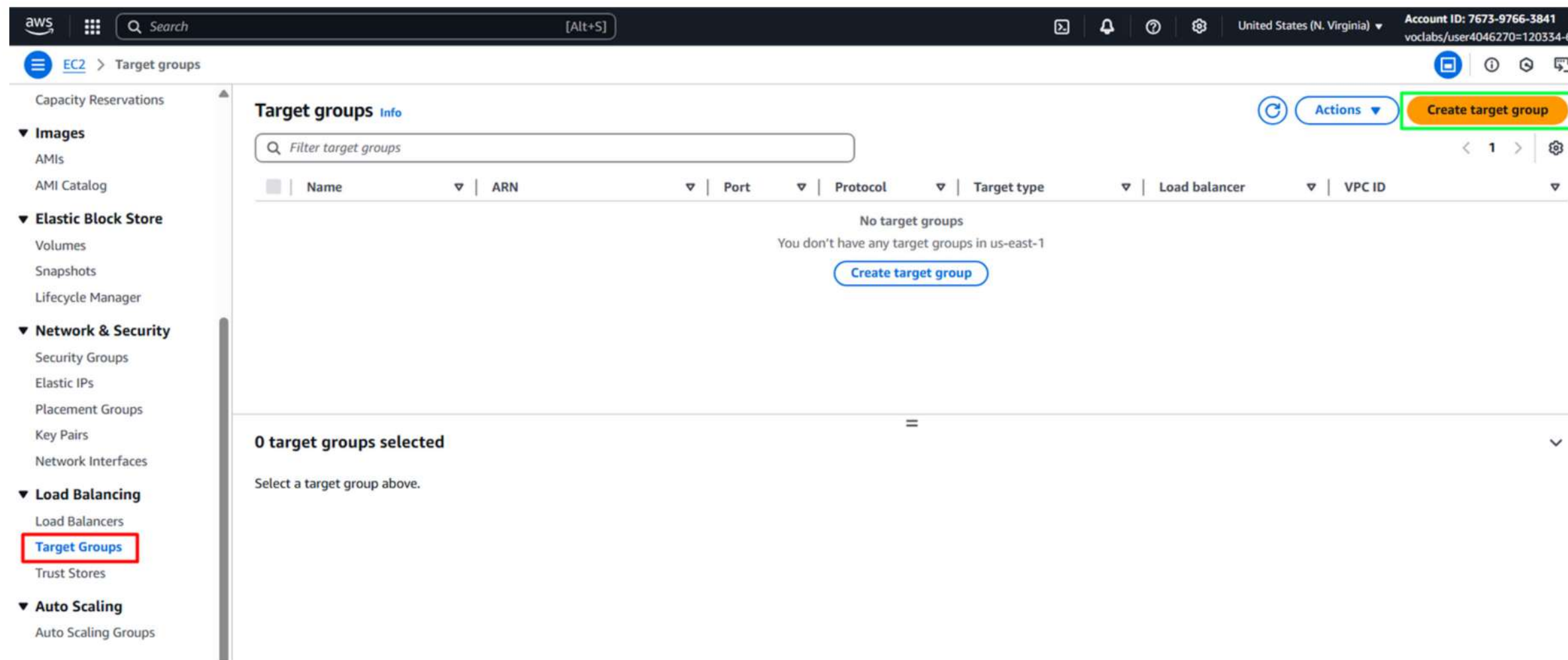
TASK [Restart Apache to apply changes] *****
changed: [192.168.1.142]

PLAY RECAP *****
192.168.1.142      : ok=21  changed=13  unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Load Balancer (LB)

Konfigurasi Load Balancer

Target group pada Load Balancer adalah kumpulan target seperti instance EC2 yang didaftarkan untuk menerima trafik dari load balancer, berdasarkan protokol dan port yang ditentukan, sehingga memudahkan pengaturan distribusi beban dan pemantauan koneksi ke setiap target.



Konfigurasi Load Balacer

Basic configuration

Settings in this section can't be changed after the target group is created.

Choose a target type

☒ Instances

- Supports load balancing to instances within a specific VPC.
- Facilitates the use of [Amazon EC2 Auto Scaling](#) to manage and scale your EC2 capacity.

☐ IP addresses

- Supports load balancing to VPC and on-premises resources.
- Facilitates routing to multiple IP addresses and network interfaces on the same instance.
- Offers flexibility with microservice based architectures, simplifying inter-application communication.
- Supports IPv6 targets, enabling end-to-end IPv6 communication, and IPv4-to-IPv6 NAT.

☐ Lambda function

- Facilitates routing to a single Lambda function.
- Accessible to Application Load Balancers only.

☐ Application Load Balancer

- Offers the flexibility for a Network Load Balancer to accept and route TCP requests within a specific VPC.
- Facilitates using static IP addresses and PrivateLink with an Application Load Balancer.

Target group name

ALB-TargetCMS

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Konfigurasi Load Balacer

Protocol : Port

Choose a protocol for your target group that corresponds to the Load Balancer type that will route traffic to it. Some protocols now include anomaly detection for the targets and you can set mitigation options once your target group is created. This choice cannot be changed after creation

HTTP

80

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

☒ IPv4

Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ IPv6

Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

VPC-Group1-vpc

vpc-012dfddf7fb5909e3

IPv4 VPC CIDR: 192.168.1.0/24

Protocol version

☒ HTTP1

Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

☐ HTTP2

Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

☐ gRPC

Send requests to targets using gRPC. Supported when the request protocol is gRPC.

Konfigurasi Load Balacer

Available instances (1/3)

Filter instances

< 1 > ⚙

☐	Instance ID	Name	State	Security groups	Zone	Private IPv
☐	i-0981f2867ef76384c	TargetBlog	Running	SG-TargetBlog	us-east-1a	192.168.1.
☒	i-04e877d36e99d8e58	TargetCMS	Running	SG-TargetCMS	us-east-1a	192.168.1.
☐	i-00f55a6c0947417a7	AnsibleNode	Running	SG-AnsibleNode	us-east-1a	192.168.1.

1 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

Include as pending below

Review targets

Targets (1)

Filter targets

Show only pending

Remove all pending

< 1 > ⚙

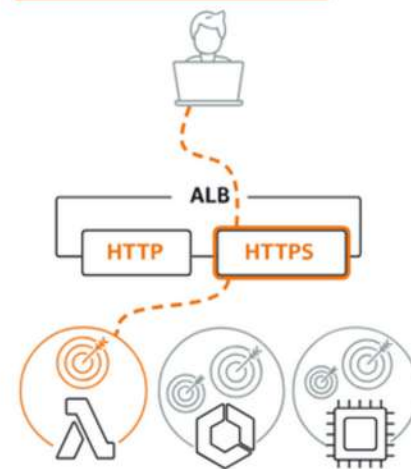
Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
i-04e877d36e99d8e58	TargetCMS	80	Running	SG-TargetCMS	us-east-1a	192.168.1.140	subnet-0fc7f8b6696225c45	May 2, 2025, 07:01 (U

Konfigurasi Load Balacer

Setelah membuat target group, langkah berikutnya adalah mengatur **Application Load Balancer (ALB)**, yaitu layanan yang secara otomatis mendistribusikan trafik masuk ke beberapa target seperti EC2 instance, berdasarkan aturan routing yang dapat disesuaikan.

Load balancer types

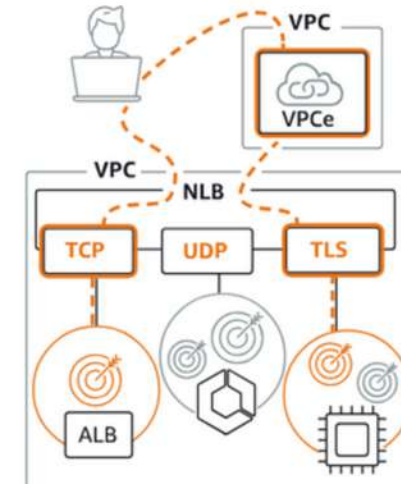
Application Load Balancer [Info](#)



Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

[Create](#)

Network Load Balancer [Info](#)



Choose a Network Load Balancer when you need ultra-high performance, TLS offloading at scale, centralized certificate deployment, support for UDP, and static IP addresses for your applications. Operating at the connection level, Network Load Balancers are capable of handling millions of requests per second securely while maintaining ultra-low latencies.

[Create](#)

Gateway Load Balancer [Info](#)



Choose a Gateway Load Balancer when you need to deploy and manage a fleet of third-party virtual appliances that support GENEVE. These appliances enable you to improve security, compliance, and policy controls.

[Create](#)

Konfigurasi Load Balacer

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

ALB-TargetCMS

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)

Scheme can't be changed after the load balancer is created.

- ☒ **Internet-facing**
- Serves internet-facing traffic.
 - Has public IP addresses.
 - DNS name resolves to public IPs.
 - Requires a public subnet.

- ☐ **Internal**
- Serves internal traffic.
 - Has private IP addresses.
 - DNS name resolves to private IPs.
 - Compatible with the **IPv4** and **Dualstack** IP address types.

Load balancer IP address type [Info](#)

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

- ☒ **IPv4**
Includes only IPv4 addresses.
- ☐ **Dualstack**
Includes IPv4 and IPv6 addresses.
- ☐ **Dualstack without public IPv4**
Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with **internet-facing** load balancers only.

Konfigurasi Load Balacer

Network mapping [Info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC [Info](#)

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#). For a new VPC, [create a VPC](#).

VPC-Group1-vpc

vpc-012dfddf7fb5909e3

IPv4 VPC CIDR: 192.168.1.0/24



IP pools - new [Info](#)

You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view **Pools** in [Amazon VPC IP Address Manager console](#).

☐ Use IPAM pool for public IPv4 addresses

The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets [Info](#)

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ **us-east-1a (use1-az2)**

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0fc7f8b6696225c45

IPv4 subnet CIDR: 192.168.1.128/28

VPC-Group1-subnet-private1-us-east-1a



The selected subnet does not have a route to an internet gateway. This means that your load balancer will not receive internet traffic.

You can proceed with this selection; however, for internet traffic to reach your load balancer, you must update the subnet's route table in the [VPC console](#).

☒ **us-east-1b (use1-az4)**

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

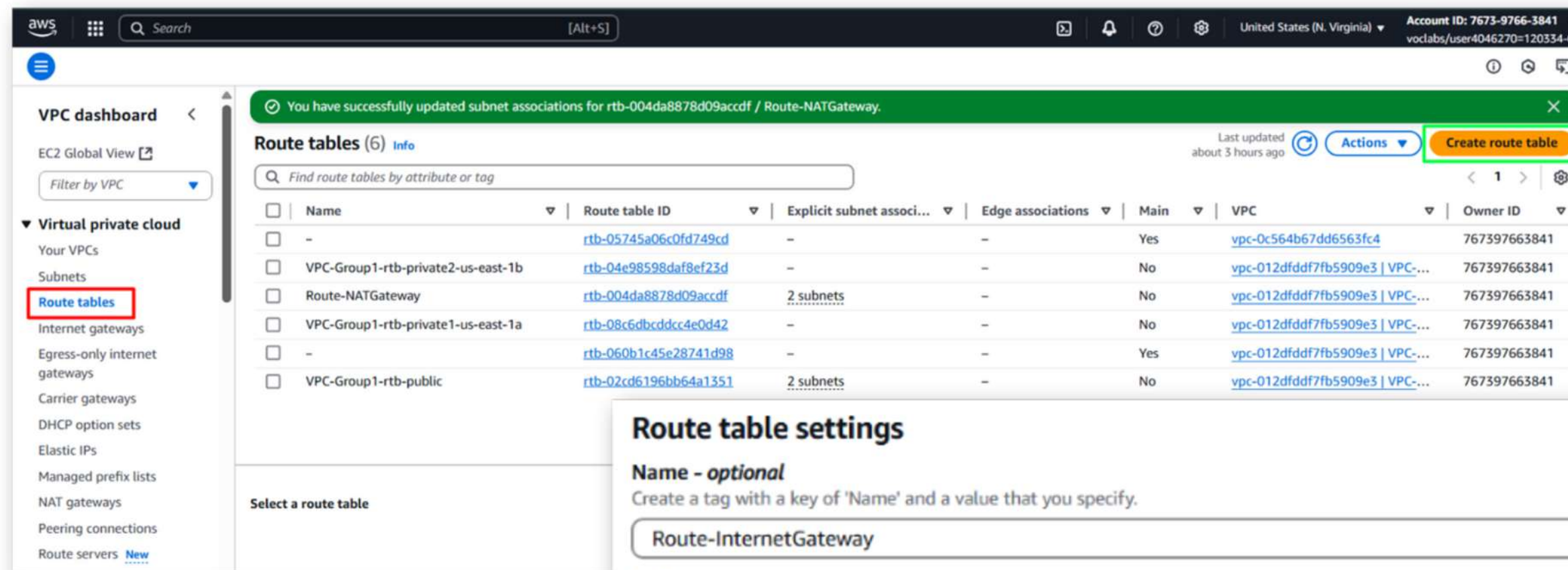
subnet-03e8d32a359f36cad

IPv4 subnet CIDR: 192.168.1.144/28

VPC-Group1-subnet-private2-us-east-1b

Konfigurasi Load Balacer

Route table IGW di subnet publik memungkinkan trafik dari internet untuk mencapai ALB. Tanpa IGW, DNS pada ALB tidak bisa diakses karena tidak ada jalur keluar-masuk trafik dari internet.



The screenshot shows the AWS Management Console interface. On the left, the 'VPC dashboard' is visible with a sidebar containing links to 'Your VPCs', 'Subnets', 'Route tables' (highlighted with a red box), 'Internet gateways', 'Egress-only internet gateways', 'Carrier gateways', 'DHCP option sets', 'Elastic IPs', 'Managed prefix lists', 'NAT gateways', 'Peering connections', and 'Route servers'. The main content area displays a notification at the top: 'You have successfully updated subnet associations for rtb-004da8878d09accdf / Route-NATGateway.' Below this, the 'Route tables (6)' section is shown with a search bar and a table of route tables. The table has columns for Name, Route table ID, Explicit subnet associations, Edge associations, Main, VPC, and Owner ID. The table lists six route tables, including 'Route-NATGateway' and 'VPC-Group1-rtb-public'. A 'Create route table' button is visible in the top right of the route tables section.

Name	Route table ID	Explicit subnet associations	Edge associations	Main	VPC	Owner ID
-	rtb-05745a06c0fd749cd	-	-	Yes	vpc-0c564b67dd6563fc4	767397663841
VPC-Group1-rtb-private2-us-east-1b	rtb-04e98598daf8ef23d	-	-	No	vpc-012dfddf7fb5909e3 VPC-...	767397663841
Route-NATGateway	rtb-004da8878d09accdf	2 subnets	-	No	vpc-012dfddf7fb5909e3 VPC-...	767397663841
VPC-Group1-rtb-private1-us-east-1a	rtb-08c6dbcdcc4e0d42	-	-	No	vpc-012dfddf7fb5909e3 VPC-...	767397663841
-	rtb-060b1c45e28741d98	-	-	Yes	vpc-012dfddf7fb5909e3 VPC-...	767397663841
VPC-Group1-rtb-public	rtb-02cd6196bb64a1351	2 subnets	-	No	vpc-012dfddf7fb5909e3 VPC-...	767397663841

Route table settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

Route-InternetGateway

VPC

The VPC to use for this route table.

vpc-012dfddf7fb5909e3 (VPC-Group1-vpc)

Konfigurasi Load Balacer

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (1)

Filter routes

Both

Edit routes

< 1 >

Destination	Target	Status	Propagated
192.168.1.0/24	local	Active	No

Edit routes

Destination

192.168.1.0/24

Target

local

local

Internet Gateway

igw-

Use: "igw-"

igw-0b7721d16a33c8233 (VPC-Group1-igw)

Status

Active

-

Propagated

No

No

Remove

Add route

Cancel

Preview

Save changes

Konfigurasi Load Balacer

Routes

Subnet associations

Edge associations

Route propagation

Tags

Explicit subnet associations (0)

Find subnet association

< 1 > ⚙

Name

Subnet ID

IPv4 CIDR

IPv6 CIDR

No subnet associations

You do not have any subnet associations.

Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/4)

Filter subnet associations

< 1 > ⚙

Name

Subnet ID

IPv4 CIDR

IPv6 CIDR

Route table ID

☒

VPC-Group1-subnet-private1-us-east-1a

subnet-0fc7f8b6696225c45

192.168.1.128/28

-

rtb-08c6dbcddcc4e0d42 / VPC-Group1...

☒

VPC-Group1-subnet-private2-us-east-1b

subnet-03e8d32a359f36cad

192.168.1.144/28

-

rtb-04e98598daf8ef23d / VPC-Group1...

☐

VPC-Group1-subnet-public1-us-east-1a

subnet-0f39347344ba40a87

192.168.1.0/28

-

rtb-02cd6196bb64a1351 / VPC-Group...

☐

VPC-Group1-subnet-public2-us-east-1b

subnet-0c95478eb60b22b5f

192.168.1.16/28

-

rtb-02cd6196bb64a1351 / VPC-Group...

Selected subnets

subnet-0fc7f8b6696225c45 / VPC-Group1-subnet-private1-us-east-1a

subnet-03e8d32a359f36cad / VPC-Group1-subnet-private2-us-east-1b

Cancel

Save associations

Konfigurasi Load Balacer

Network mapping [Info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC [Info](#)

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#). For a new VPC, [create a VPC](#).

VPC-Group1-vpc
vpc-012dfddf7fb5909e3
IPv4 VPC CIDR: 192.168.1.0/24



IP pools - new [Info](#)

You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view **Pools** in [Amazon VPC IP Address Manager console](#).

☐ Use IPAM pool for public IPv4 addresses

The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets [Info](#)

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ **us-east-1a (use1-az2)**

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0fc7f8b6696225c45
IPv4 subnet CIDR: 192.168.1.128/28

VPC-Group1-subnet-private1-us-east-1a

☒ **us-east-1b (use1-az4)**

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-03e8d32a359f36cad
IPv4 subnet CIDR: 192.168.1.144/28

VPC-Group1-subnet-private2-us-east-1b

Konfigurasi Load Balacer

Load balancers (2)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

< 1 > ⚙

☐	Name	DNS name	State	VPC ID	Availability Zones	Type	Date created
☐	ALB-TargetCMS	ALB-TargetCMS-864384365...	⋯ Provisioning	vpc-012dfddf7fb5909e3	2 Availability Zones	application	May 2, 2025, 09:55 (UTC+07:00)
☐	ALB-TargetBlog	ALB-TargetBlog-11751331....	⋯ Provisioning	vpc-012dfddf7fb5909e3	2 Availability Zones	application	May 2, 2025, 09:57 (UTC+07:00)

Load balancers (2)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

< 1 > ⚙

☐	Name	DNS name	State	VPC ID	Availability Zones	Type	Date created
☐	ALB-TargetCMS	ALB-TargetCMS-864384365...	✅ Active	vpc-012dfddf7fb5909e3	2 Availability Zones	application	May 2, 2025, 09:55 (UTC+07:00)
☐	ALB-TargetBlog	ALB-TargetBlog-11751331....	✅ Active	vpc-012dfddf7fb5909e3	2 Availability Zones	application	May 2, 2025, 09:57 (UTC+07:00)

Konfigurasi Load Balacer

WordPress - Installation

Not secure alb-targetcms-864384365.us-east-1.elb.amazonaws.com/wp-admin/install.php

Welcome

Welcome to the famous five-minute WordPress installation process! Just fill in the information below and you'll be on your way to using the most extendable and powerful personal publishing platform in the world.

Information needed

Please provide the following information. Do not worry, you can always change these settings later.

Site Title: UKK 2025

Username: wordpress

Password: wordpress (Very weak)

Confirm Password: ☒ Confirm use of weak password

Your Email: dar.haidaryahya@gmail.com

Search engine visibility: ☐ Discourage search engines from indexing this site

[Install WordPress](#)

WordPress - Installation

Not secure alb-targetblog-11751331.us-east-1.elb.amazonaws.com

My Blog

[Home](#) | [Guestbook](#)

Cari posting: [Cari](#)

[Bendless Love](#)

Tanggal 2013-05-29 00:00:00

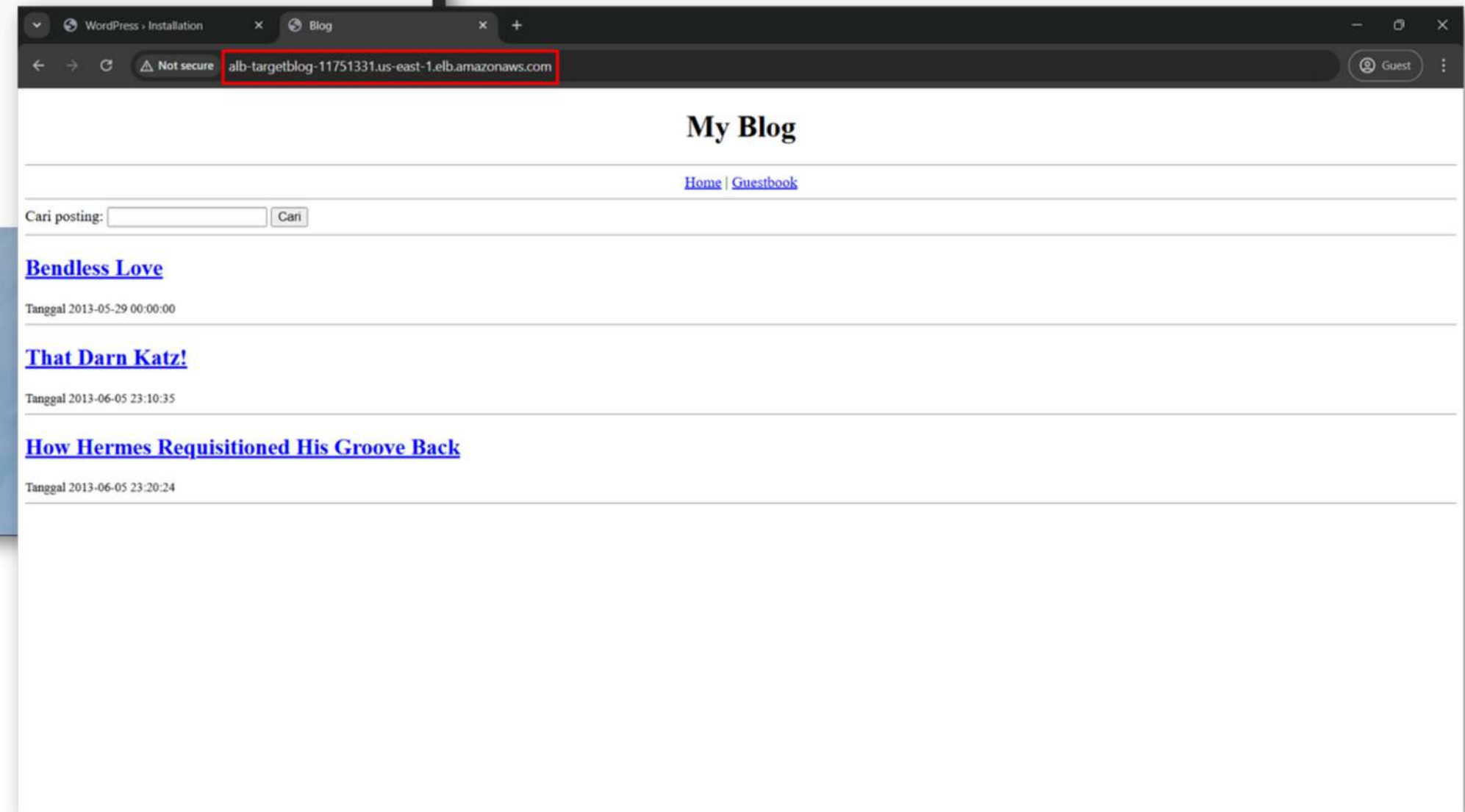
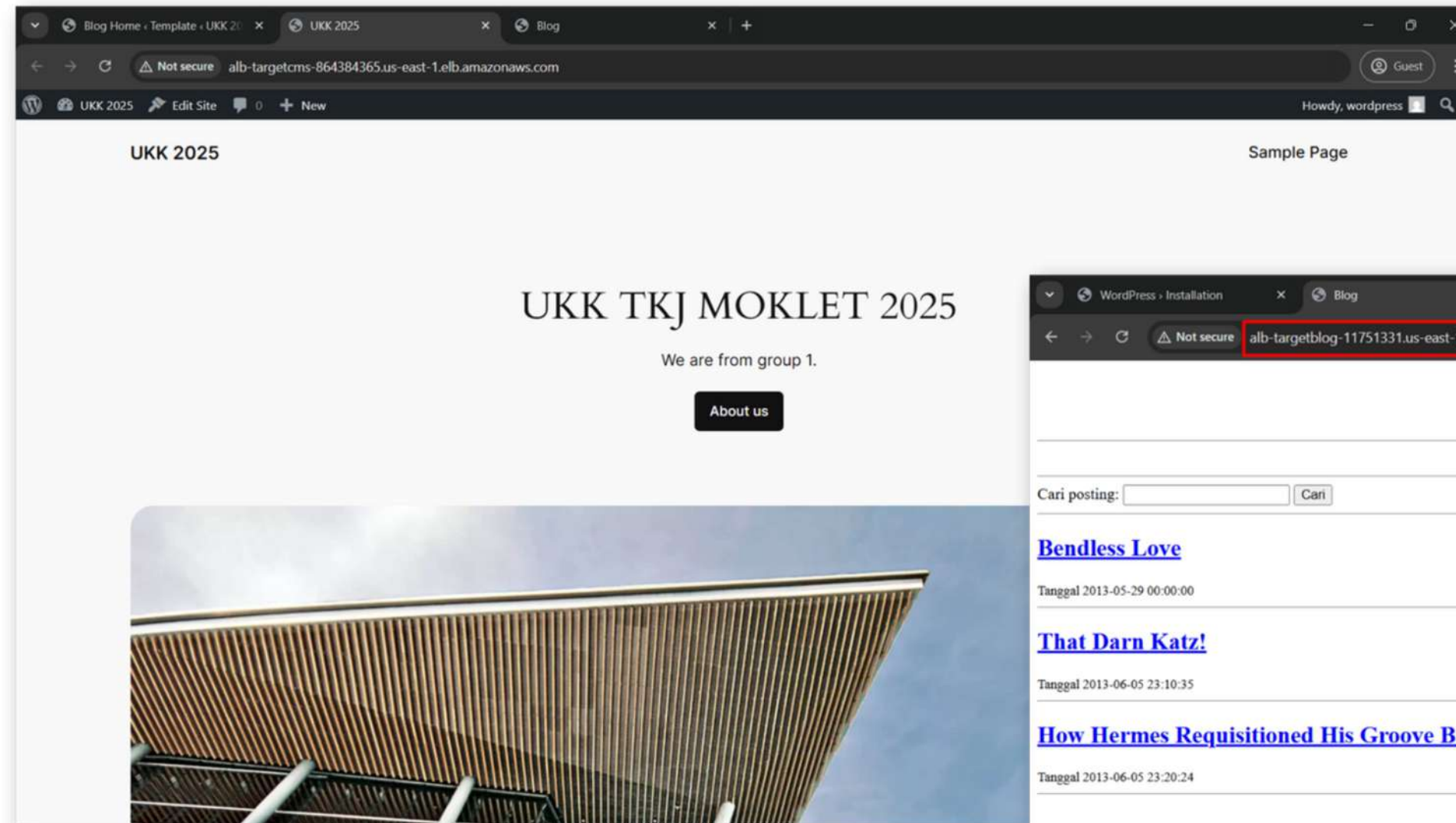
[That Darn Katz!](#)

Tanggal 2013-06-05 23:10:35

[How Hermes Requisitioned His Groove Back](#)

Tanggal 2013-06-05 23:20:24

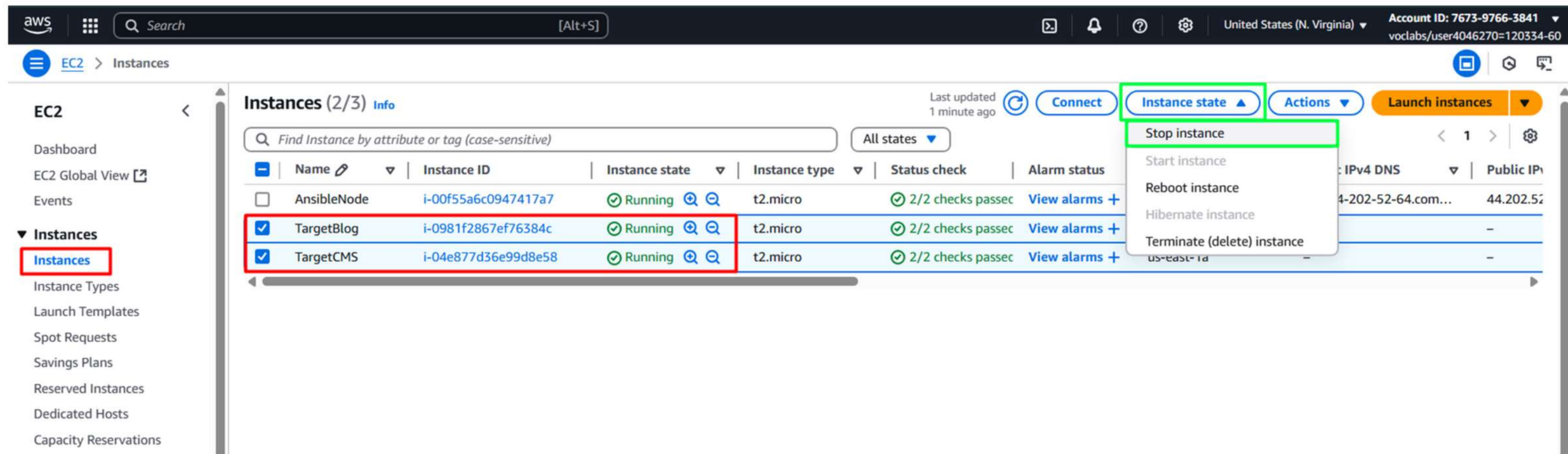
Konfigurasi Load Balacer



Auto Scaling Groups (ASG)

Konfigurasi Auto Scaling Groups

Setelah mengatur ALB, selanjutnya adalah membuat **Auto Scaling Group (ASG)**. Berfungsi untuk secara otomatis menambah atau mengurangi jumlah instance berdasarkan kebutuhan trafik. Ini memastikan aplikasi tetap dapat menangani beban yang meningkat atau menurunkan biaya saat beban berkurang.

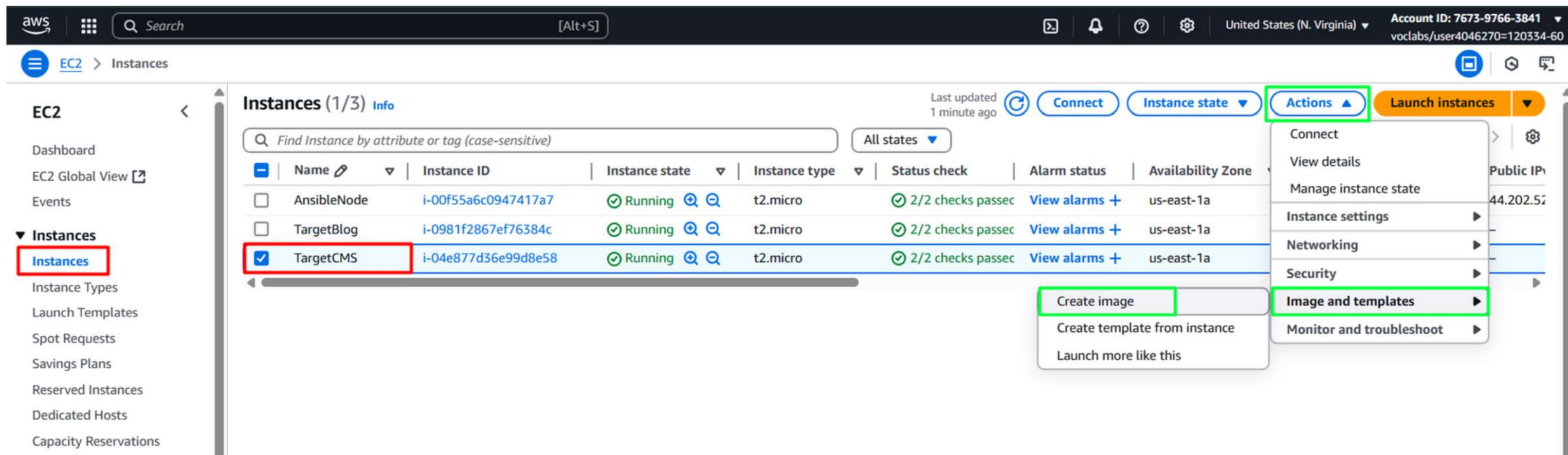


The screenshot displays the AWS Management Console interface for EC2 instances. The left sidebar shows the navigation menu with 'Instances' highlighted. The main panel shows a table of instances. The 'TargetBlog' and 'TargetCMS' instances are highlighted with a red box. The 'Instance state' dropdown menu is open, showing options like 'Stop instance', 'Start instance', 'Reboot instance', 'Hibernate instance', and 'Terminate (delete) instance'.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status
AnsibleNode	i-00f55a6c0947417a7	Running	t2.micro	2/2 checks passed	View alarms
TargetBlog	i-0981f2867ef76384c	Running	t2.micro	2/2 checks passed	View alarms
TargetCMS	i-04e877d36e99d8e58	Running	t2.micro	2/2 checks passed	View alarms

Konfigurasi Auto Scaling Groups

Sebelum membuat Auto Scaling Group (ASG), siapkan terlebih dahulu **image AMI** dan **template instance**. **AMI** digunakan untuk membuat salinan instance yang akan digunakan oleh ASG, sementara **template instance** berisi konfigurasi dasar seperti jenis instance, security group, dan lainnya, yang diperlukan untuk membuat instance baru dalam ASG.



The screenshot displays the AWS Management Console interface for the EC2 Instances page. The left sidebar shows the navigation menu with 'Instances' highlighted. The main content area shows a table of instances. The 'TargetCMS' instance is selected, and the 'Actions' dropdown menu is open, showing options like 'Create image', 'Create template from instance', and 'Launch more like this'. The 'Image and templates' option is highlighted.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
AnsibleNode	i-00f55a6c0947417a7	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a
TargetBlog	i-0981f2867ef76384c	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a
TargetCMS	i-04e877d36e99d8e58	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a

Konfigurasi Auto Scaling Groups

Instance ID
i-04e877d36e99d8e58 (TargetCMS)

Image name
Image-TargetCMS
Maximum 127 characters. Can't be modified after creation.

Image description - optional
Image for TargetCMS
Maximum 255 characters

☒ **Reboot instance**
When selected, Amazon EC2 reboots the instance so that data is at rest when snapshots of the attached volumes are taken. This ensures data consistency.

Instance volumes

Storage type	Device	Snapshot
EBS	/dev/s...	Create new

[Add volume](#)

aws Search [Alt+S]

United States (N. Virginia) Account ID: 7673-9766-3841
voclabs/user4046270=120334-60

EC2 > AMIs

Amazon Machine Images (AMIs) (2) Info

Owned by me Find AMI by attribute or tag

	Name	AMI name	AMI ID	Source	Owner	Visibility	Status
☐		Image-TargetBlog	ami-06ac3c97280b2a678	767397663841/Image-TargetBlog	767397663841	Private	Available
☐		Image-TargetCMS	ami-070722c607ff20fa1	767397663841/Image-TargetCMS	767397663841	Private	Available

Select an AMI

Konfigurasi Auto Scaling Groups

The screenshot displays the AWS Management Console interface. At the top, the navigation bar includes the AWS logo, a search bar, and account information for 'United States (N. Virginia)' with Account ID 7673-9766-3841. The left-hand navigation pane is expanded to the 'EC2' section, where 'Launch Templates' is highlighted with a red rectangular box. The main content area, titled 'Compute', features a large heading 'EC2 launch templates' followed by the subtext 'Streamline, simplify and standardize instance launches'. Below this, a paragraph explains the utility of launch templates for automating instance launches and enforcing best practices. A 'New launch template' call-to-action box is positioned on the right, containing an orange 'Create launch template' button that is outlined with a green border. At the bottom of the page, the text 'Benefits and features' is visible.

aws [Search] [Alt+S] United States (N. Virginia) Account ID: 7673-9766-3841
voclabs/user4046270=120334-60

EC2

- Dashboard
- EC2 Global View
- Events
- ▼ **Instances**
 - Instances
 - Instance Types
 - Launch Templates**
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
- ▼ **Images**
 - AMIs
 - AMI Catalog

Compute

EC2 launch templates

Streamline, simplify and standardize instance launches

Use launch templates to automate instance launches, simplify permission policies, and enforce best practices across your organization. Save launch parameters in a template that can be used for on-demand launches and with managed services, including EC2 Auto Scaling and EC2 Fleet. Easily update your launch parameters by creating a new launch template version.

New launch template

Create launch template

Benefits and features

Konfigurasi Auto Scaling Groups

Launch template name and description

Launch template name - *required*

Template-TargetCMS

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '*', '@'.

Template version description

Template for TargetCMS

Max 255 chars

Auto Scaling guidance | [Info](#)

Select this if you intend to use this template with EC2 Auto Scaling

☒ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

- ▶ Template tags
- ▶ Source template

Konfigurasi Auto Scaling Groups

▼ **Application and OS Images (Amazon Machine Image) - required** [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Recents

My AMIs

Quick Start

☒ Owned by me

☐ Shared with me



[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)	
Image-TargetCMS ami-070722c607ff20fa1 2025-05-02T03:20:55.000Z Virtualization: hvm ENA enabled: true Root device type: ebs Boot mode: uefi-preferred	
Description	Image for TargetCMS
Architecture	AMI ID
x86_64	ami-070722c607ff20fa1

Konfigurasi Auto Scaling Groups

▼ Instance type [Info](#) | [Get advice](#)

Advanced

Instance type

t2.micro

Free tier eligible

Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.0162 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour On-Demand SUSE base pricing: 0.0116 USD per Hour
On-Demand RHEL base pricing: 0.026 USD per Hour On-Demand Linux base pricing: 0.0116 USD per Hour

☐ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

vockey

 [Create new key pair](#)

Konfigurasi Auto Scaling Groups

▼ Network settings [Info](#)

Subnet [Info](#)

Don't include in launch template



[Create new subnet](#)

When you specify a subnet, a network interface is automatically added to your template.

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Select existing security group

☐ Create security group

Security groups [Info](#)

Select security groups



SG-TargetCMS sg-010485e863cd23e04 ✕
VPC: vpc-012dfddf7fb5909e3



[Compare security group rules](#)

► Advanced network configuration

Konfigurasi Auto Scaling Groups

The screenshot shows the AWS Management Console interface for the 'Launch Templates' page. The left sidebar shows the 'EC2' section with 'Launch Templates' highlighted. The main content area displays a table of launch templates. The first row, 'Template-TargetCMS' with ID 'lt-070c884bfddb2f332', is selected. The 'Actions' menu is open, showing options like 'Launch instance from template', 'Modify template (Create new version)', 'Delete template', 'Delete template version', 'Set default version', 'Manage tags', 'Create Spot Fleet', 'Create Auto Scaling group', and 'View details'. The 'Create Auto Scaling group' option is highlighted. The 'Launch Template Name' 'Template-TargetCMS' is also highlighted.

Launch Templates (1/2)

Launch Template ID	Launch Template Name	Default Version	Latest Version	Create Date
☒ lt-070c884bfddb2f332	Template-TargetCMS	1	1	2020-10-20 10:10:10
☐ lt-07bbe3c610051b4b6	Template-TargetBlog	1	1	2020-10-20 10:10:10

Template-TargetCMS (lt-070c884bfddb2f332)

Konfigurasi Auto Scaling Groups

Name

Auto Scaling group name

Enter a name to identify the group.

ASG-TargetCMS

Must be unique to this account in the current Region and no more than 255 characters.

Launch template [Info](#)

i For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template

Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

Template-TargetCMS

[Create a launch template](#)

Version

Default (1)

[Create a launch template version](#)

Description

Template for TargetCMS

AMI ID

ami-070722c607ff20fa1

Key pair name

vockey

Launch template

[Template-TargetCMS](#)

lt-070c884bfddb2f332

Security groups

-

Security group IDs

[sg-010485e863cd23e04](#)

Instance type

t2.micro

Request Spot Instances

No

Konfigurasi Auto Scaling Groups

Network [Info](#)

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC

Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-012dfddf7fb5909e3 (VPC-Group1-vpc)
192.168.1.0/24



[Create a VPC](#)

Availability Zones and subnets

Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets



us-east-1b | subnet-03e8d32a359f36cad (VPC-Group1-subnet-private2-us-east-1b) X
192.168.1.144/28

[Create a subnet](#)

Availability Zone distribution - new

Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

☒ **Balanced best effort**
If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.

☐ **Balanced only**
If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy Availability Zone to preserve balanced distribution.

Konfigurasi Auto Scaling Groups

Load balancing [Info](#)

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

- ☐ No load balancer
Traffic to your Auto Scaling group will not be fronted by a load balancer.
- ☒ Attach to an existing load balancer
Choose from your existing load balancers.
- ☐ Attach to a new load balancer
Quickly create a basic load balancer to attach to your Auto Scaling group.

Attach to an existing load balancer

Select the load balancers that you want to attach to your Auto Scaling group.

- ☒ Choose from your load balancer target groups
This option allows you to attach Application, Network, or Gateway Load Balancers.
- ☐ Choose from Classic Load Balancers

Existing load balancer target groups

Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.

Select target groups



ALB-TargetCMS | HTTP 
Application Load Balancer: ALB-TargetCMS

Konfigurasi Auto Scaling Groups

Scaling [Info](#)

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

Equal or less than desired capacity

Max desired capacity

Equal or greater than desired capacity

Automatic scaling - *optional*

Choose whether to use a target tracking policy | [Info](#)

You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☒ No scaling policies

Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☐ Target tracking scaling policy

Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Konfigurasi Auto Scaling Groups

Instances (2/3) Info

Find Instance by attribute or tag (case-sensitive)

All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status
AnsibleNode	i-00f55a6c0947417a7	Running	t2.micro	2/2 checks passed	View alarms
TargetBlog	i-0981f2867ef76384c	Running	t2.micro	2/2 checks passed	View alarms
TargetCMS	i-04e877d36e99d8e58	Running	t2.micro	2/2 checks passed	View alarms

Instance state dropdown menu:

- Stop instance
- Start instance
- Reboot instance
- Hibernate instance
- Terminate (delete) instance

Successfully initiated stopping of i-04e877d36e99d8e58, i-0981f2867ef76384c

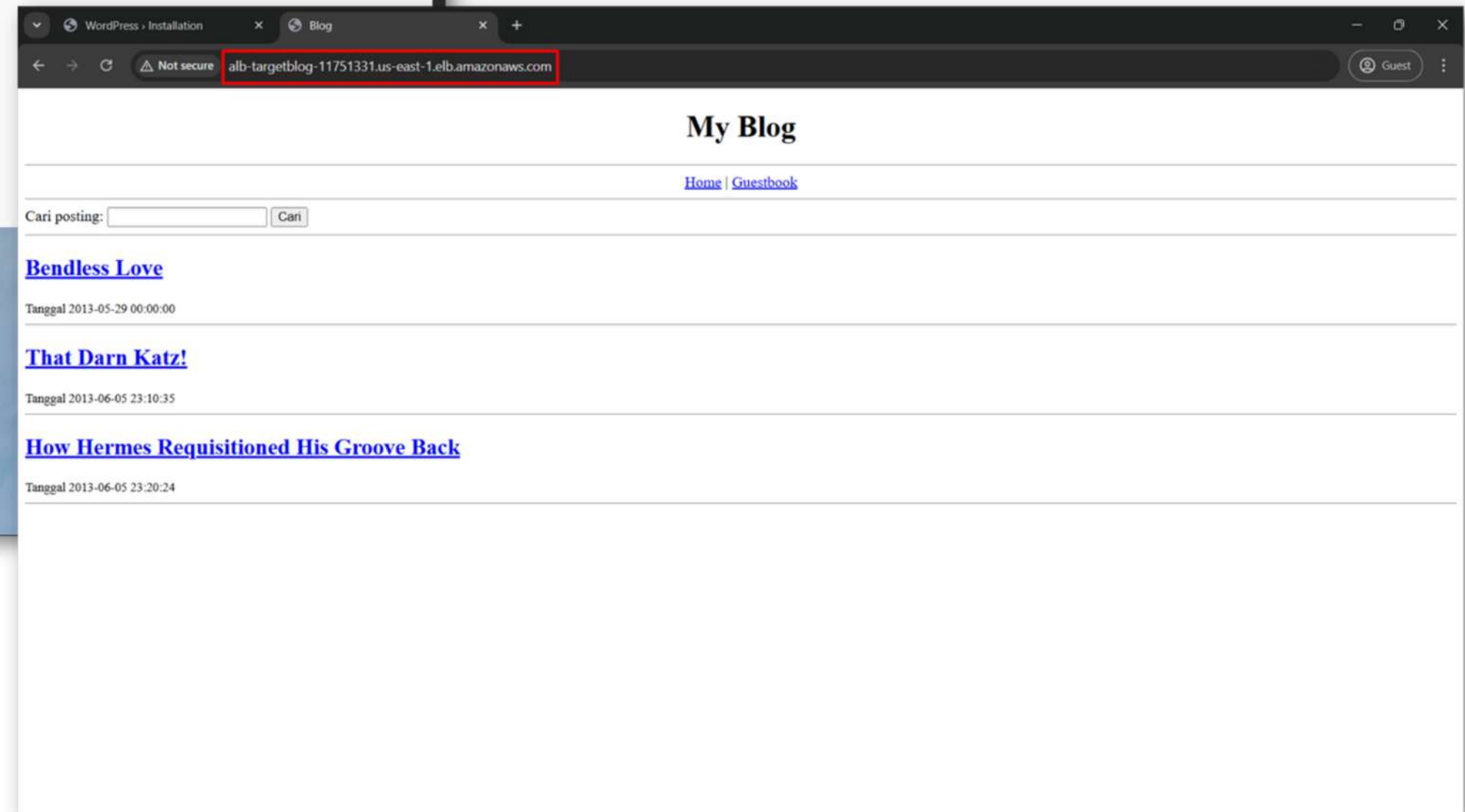
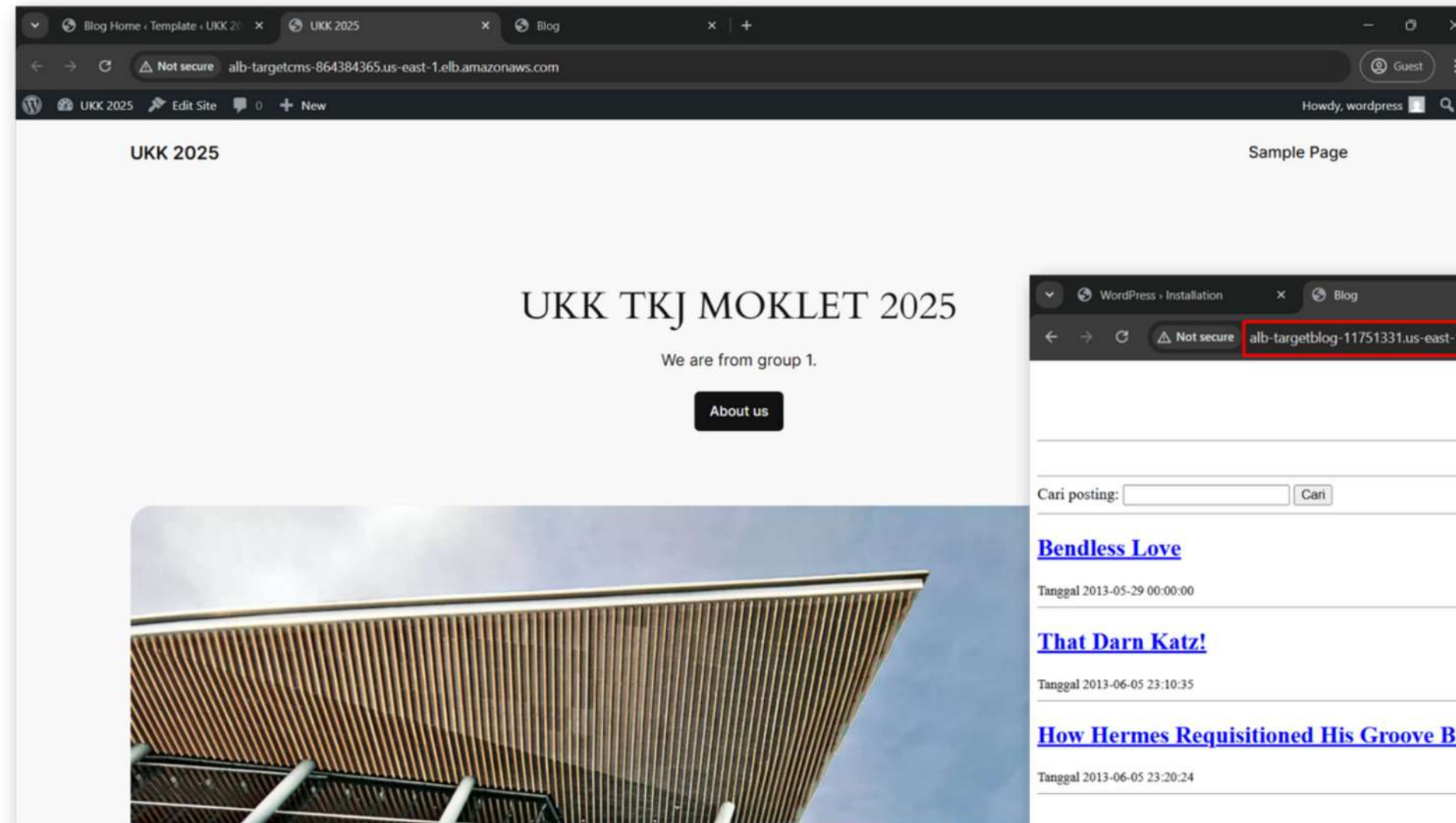
Instances (5) Info

Find Instance by attribute or tag (case-sensitive)

All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IP
	i-0bfeac57545ffe97c	Running	t2.micro	2/2 checks passed	View alarms	us-east-1b	-	-
	i-090f6c50d24e40540	Running	t2.micro	2/2 checks passed	View alarms	us-east-1b	-	-
AnsibleNode	i-00f55a6c0947417a7	Running	t2.micro	2/2 checks passed	View alarms	us-east-1a	ec2-44-202-52-64.com...	44.202.52...
TargetBlog	i-0981f2867ef76384c	Stopping	t2.micro	-	View alarms	us-east-1a	-	-
TargetCMS	i-04e877d36e99d8e58	Stopped	t2.micro	-	View alarms	us-east-1a	-	-

Konfigurasi Auto Scaling Groups



VulnWeb Patching

VulnWeb Patching

Langkah awal patching dimulai dengan melakukan pemindaian menggunakan **tool dirb** untuk mendeteksi direktori tersembunyi. Hasil scan menemukan file **admin.php**, yang kemudian dianalisis lebih lanjut sebagai celah yang perlu diamankan.

```
root@ip-192-168-1-7:/home/ubuntu# apt install dirb -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  dirb
0 upgraded, 1 newly installed, 0 to remove and 0 not installed.
```

```
root@ip-192-168-1-7:/home/ubuntu# dirb http://alb-targetblog-11751331.us-east-1.elb.amazonaws.com/
-----
DIRB v2.22
By The Dark Raver
-----

START_TIME: Fri May  2 15:56:47 2025
URL_BASE: http://alb-targetblog-11751331.us-east-1.elb.amazonaws.com/
WORDLIST_FILES: /usr/share/dirb/wordlists/common.txt

-----

GENERATED WORDS: 4612

---- Scanning URL: http://alb-targetblog-11751331.us-east-1.elb.amazonaws.com/ ----
+ http://alb-targetblog-11751331.us-east-1.elb.amazonaws.com/.git/HEAD (CODE:200|SIZE:21)
+ http://alb-targetblog-11751331.us-east-1.elb.amazonaws.com/admin.php (CODE:302|SIZE:382)
+ http://alb-targetblog-11751331.us-east-1.elb.amazonaws.com/index.php (CODE:200|SIZE:805)
+ http://alb-targetblog-11751331.us-east-1.elb.amazonaws.com/server-status (CODE:403|SIZE:316)

-----

END_TIME: Fri May  2 15:56:52 2025
DOWNLOADED: 4612 - FOUND: 4
```

VulnWeb Patching

Setelah menemukan *admin.php*, kita masuk ke *server targetblog* via SSH dan membuka file tersebut. Di dalamnya terdapat informasi redirect ke *admin_login.php*, yang mengindikasikan lokasi login admin dan menjadi fokus untuk patching selanjutnya.

```
ubuntu@ip-192-168-1-7:~$ ssh -i "vockey.pem" ubuntu@192.168.1.142
The authenticity of host '192.168.1.142 (192.168.1.142)' can't be established.
ED25519 key fingerprint is SHA256:bRjTEHyKD+N4zzxBGi2dxa5DA5p+1V4p6rknbP4mZy8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.142' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.8.0-1024-aws x86_64)
```

```
ubuntu@ip-192-168-1-142:~$ sudo su
root@ip-192-168-1-142:/home/ubuntu# cd /var/www/html
root@ip-192-168-1-142:/var/www/html# ls
admin.php  admin_login.php  admin_logout.php  db.sql  gb.php  gb_post.php  index.php  koneksi.php  post.php  search.php
root@ip-192-168-1-142:/var/www/html# ll
total 56
drwxr-xr-x 3 root root 4096 May  2 01:43 ./
drwxr-xr-x 3 root root 4096 May  2 01:43 ../
drwxr-xr-x 8 root root 4096 May  2 01:43 .git/
-rw-r--r-- 1 root root  477 May  2 01:43 admin.php
-rw-r--r-- 1 root root 1060 May  2 01:43 admin_login.php
-rw-r--r-- 1 root root   71 May  2 01:43 admin_logout.php
-rw-r--r-- 1 root root 6236 May  2 01:43 db.sql
-rw-r--r-- 1 root root 1044 May  2 01:43 gb.php
-rw-r--r-- 1 root root  361 May  2 01:43 gb_post.php
-rw-r--r-- 1 root root   92 May  2 01:43 index.php
-rw-r--r-- 1 root root  160 May  2 01:43 koneksi.php
-rw-r--r-- 1 root root  869 May  2 01:43 post.php
-rw-r--r-- 1 root root 1017 May  2 01:43 search.php
root@ip-192-168-1-142:/var/www/html#
```

```
<?php

session_start();

if (!$_SESSION['admin']) {
    header("Location: admin_login.php");
}

?><!DOCTYPE html>
```

VulnWeb Patching

Setelah membuka admin_login.php, ditemukan celah SQL Injection pada baris 7 hingga 12. Di bagian ini, kode memungkinkan input seperti ' **OR 1=1 -- -**' untuk bypass login, sehingga perlu diperbaiki agar validasi input lebih aman dan tidak langsung menjalankan query dari user.

```
root@ip-192-168-1-142:/var/www/html# nano admin_login.php
```

```
GNU nano 7.2 admin_login.php
<?php
session_start();
include 'koneksi.php';

if (isset($_POST['submit'])) {
    $username = mysqli_real_escape_string($conn, $_POST['username']);
    $password = $_POST['password'];

    $login = mysqli_query($conn, "SELECT * FROM user WHERE password = '$password' AND username = '$username' ");

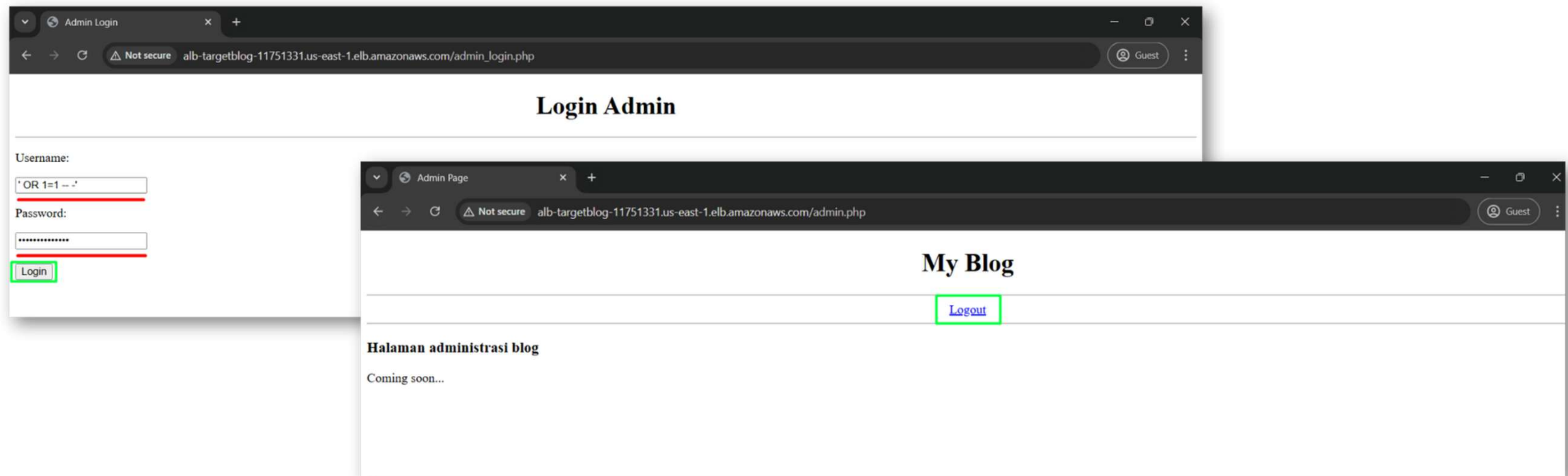
    if (mysqli_num_rows($login) == 0) {
        die("Username atau password salah!");
    } else {
        $_SESSION['admin'] = 1;
        header("Location: admin.php");
    }
}

//tambahkan atau perbaiki script diantara dibaris 7 sampai 12
//sql code injection di login admin bagian pasword ' OR 1=1 -- -'

?><!DOCTYPE html>
<html lang="en">
```

Attack

Setelah mengetahui celah SQL Injection, dilakukan pengujian serangan ke halaman login admin menggunakan payload '**OR 1=1 -- -**', dan hasilnya berhasil masuk tanpa autentikasi, membuktikan bahwa sistem rentan dan perlu segera diperbaiki.



Defend

Pada potongan kode ini terdapat celah SQL Injection karena variabel *\$password* dimasukkan langsung ke dalam pernyataan SQL tanpa dilakukan sanitasi atau escaping, sehingga penyerang dapat menyisipkan kode SQL berbahaya melalui form input dan memperoleh akses tidak sah ke dalam sistem.

```
session_start();
include 'koneksi.php';

if (isset($_POST['submit'])) {

    $username = mysqli_real_escape_string($conn, $_POST['username']);
    $password = $_POST['password'];

    $login = mysqli_query($conn, "SELECT * FROM user WHERE password = '$password' AND username = '$username' ");

    if (mysqli_num_rows($login) == 0) {
        die("Username atau password salah!");
    } else {
        $_SESSION['admin'] = 1;
        header("Location: admin.php");
    }
}
```

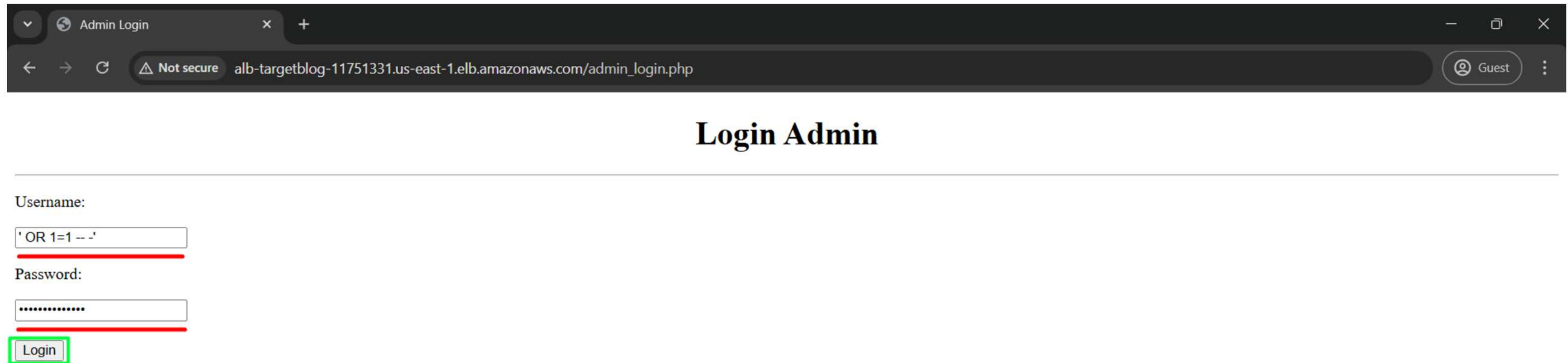
Defend

- **Kode pertama** rentan SQL Injection karena membangun query langsung dari input tanpa perlindungan.
- **Kode kedua** aman karena menggunakan teknik prepared statement untuk memisahkan perintah SQL dari data input pengguna.

```
if (isset($_POST['submit'])) {  
    /*$username = mysqli_real_escape_string($conn, $_POST['username']);  
    $password = $_POST['password'];  
  
    $login = mysqli_query($conn, "SELECT * FROM user WHERE password = '$password' AND username = '$username' ");  
  
    if (mysqli_num_rows($login) == 0) {  
        die("Username atau password salah!");  
    } else {  
        $_SESSION['admin'] = 1;  
        header("Location: admin.php");  
    }  
    */  
  
    $username = mysqli_real_escape_string($conn, $_POST['username']);  
    $password = mysqli_real_escape_string($conn, $_POST['password']);  
  
    $login = $conn->prepare("SELECT * FROM user WHERE password = ? AND username = ?");  
    $login->bind_param('ss', $password, $username);  
    $login->execute();  
    $result = $login->get_result();  
  
    if ($result->num_rows == 0) {  
        die("Username atau password salah!");  
    } else {  
        $_SESSION['admin'] = 1;  
        header("Location: admin.php");  
    }  
}
```

Defend

Setelah diperbaiki dengan prepared statement, login admin tidak bisa lagi ditembus oleh SQL Injection karena input pengguna tidak lagi dieksekusi sebagai bagian dari query, melainkan diproses sebagai data biasa sehingga celah keamanannya tertutup.



Admin Login

Not secure alb-targetblog-11751331.us-east-1.elb.amazonaws.com/admin_login.php

Login Admin

Username:

Password:

Login

The background features several large, overlapping organic shapes in shades of purple and blue. The central shape is a large, rounded circle with a gradient from light blue at the top to dark purple at the bottom. Other shapes of varying sizes and colors are scattered around it, creating a layered, abstract effect.

**Apakah Ada
Pertanyaan?**

The background features several large, overlapping, organic shapes in shades of purple and blue. These shapes have a soft, gradient-like appearance, giving them a three-dimensional, bubble-like quality. The colors range from deep indigo to a lighter, almost cyan blue, creating a vibrant and modern aesthetic.

Terima Kasih